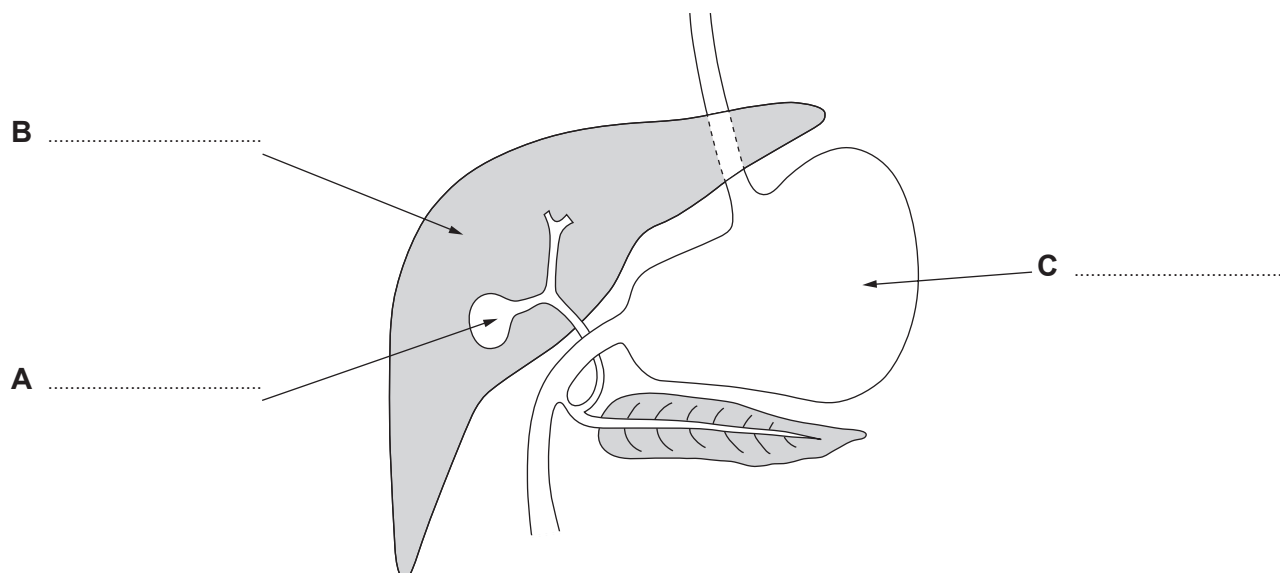


4. Food provides energy for the body. The digestive system is adapted to break down and absorb food. Parts of the digestive system are responsible for the digestion of foods by specific enzymes. The pH varies in these parts to aid the action of these enzymes in the digestion of the food.

(a) The diagram below shows part of the digestive system.



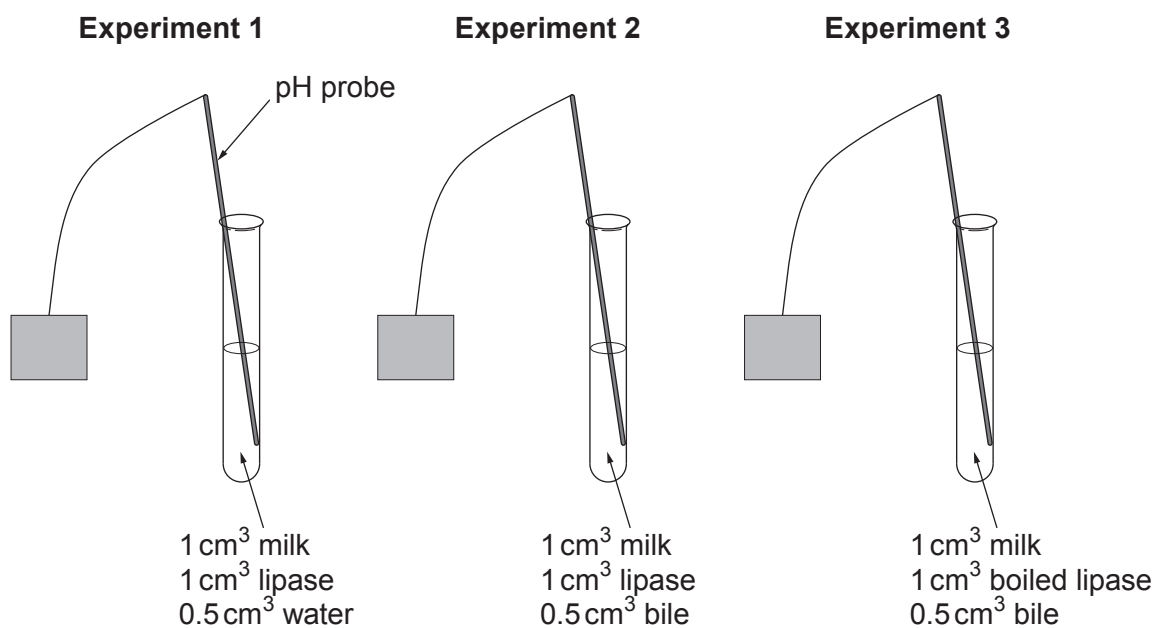
- (i) Label parts **A**, **B** and **C**. [3]
- (ii) Name the organ that secretes the enzyme lipase. [1]

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- (b) A group of students was investigating the digestion of milk fat by the enzyme lipase. Lipase has an optimum pH of 10.

The students set up three experiments.

- In each experiment they mixed an alkaline solution of milk with lipase and either bile or water.
- They used the same volume of milk, lipase and bile or water in each experiment.
- They recorded the pH after mixing for 5, 10 and 20 minutes.



In each experiment the pH was recorded. The results of the experiments are shown in the table below.

Time (minutes)	pH		
	Experiment 1	Experiment 2	Experiment 3
0	10	10	10
5	10	8	10
10	9	7	10
20	8	6	10

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only

(i) Explain the pH changes in experiments **1** and **2**.

[4]

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(ii) Explain the pH recorded in experiment **3**.

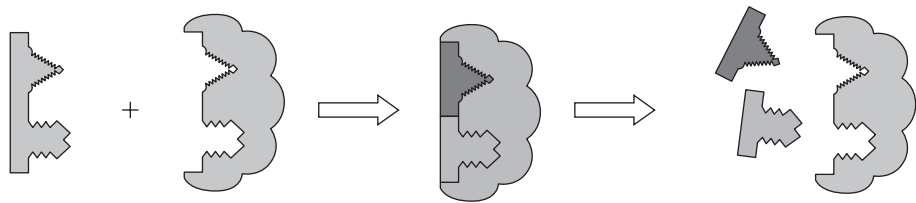
[2]

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(c) The diagram below shows a model used to explain enzyme action.



Explain how this model works in relation to the digestion of lipids.

[3]

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